Kähler Unified Transformer v2:

A Language Model Architecture Derived from a Single Geometric Potential

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Abstract

We present the Kähler Unified Transformer (KUT), a language-model architecture whose principal computation paths are modulated by quantities derived from a single Kähler potential

$$K(z, \bar{z}) = \frac{\alpha}{2}|z|^4 + \beta|z|^2.$$

Unlike a standard Transformer, whose attention, residual, normalization, feed-forward, and output-readout components are introduced as separate design choices, KUT ties these components to a shared geometric source. The metric $g = \partial_z \partial_{\bar{z}} K$ controls normalization and covariant feed-forward scaling; the connection proxy $\Gamma = 2\alpha/g$ controls attention-frame rotation and residual transport; the curvature proxy $\kappa = 2\alpha\beta/g^2$ enters the field dynamics; and the frequency $\omega = \sqrt{g_{\text{vac}}/v^2}$ defines the readout temperature $\hbar_{\text{eff}} = \pi/\omega$. We formulate a flat-geometric limit showing that, after closing the explicitly geometric side channels, KUT reduces to a Transformer-family architecture with standard dot-product attention, ordinary residual addition, constant normalization scale, and fixed output temperature. Empirically, in current KUM-Lumina-0.6B runs, the geometric parameter α increases during training, coherent greedy generations are observed, and the section readout is operationally necessary for nondegenerate output. We phrase these results as architectural validation rather than state-of-the-art language-model performance: KUT remains below fully pretrained models of comparable scale, but provides evidence that a functional language model can be built around a single learned geometric potential.

1 Introduction

Modern language models combine components that originated from separate research threads: scaled dot-product attention [1], RMS normalization [2], rotary positional embeddings [4], and gated feed-forward blocks. These parts work well together, but the standard architecture does not derive them from a common mathematical object.

This paper asks whether a functional language-model architecture can be organized around one geometric generator. We study a quartic Kähler potential

$$K(z, \bar{z}) = \frac{\alpha}{2}|z|^4 + \beta|z|^2, \tag{1}$$

where $z = z_a + iz_b \in \mathbb{C}^R$ is a learned complex field associated with each token position. The potential simultaneously induces four computational roles:

(i) **Geometry:** metric, connection proxy, curvature proxy, and vacuum scale;

(ii) **Probability:** a geometry-derived temperature $\hbar_{\text{eff}} = \pi/\omega$ and a metric amplitude in the section readout;

(iii) **Physics:** field evolution under symmetry-restoring and curvature-uniformizing forces;

(iv) **Information:** section measurement, causal section flow, logits, and cross-entropy.

The term "derived" is used in an operational sense: the implemented modulation scalars are explicit algebraic functions of equation (1) and its derivatives or derivative proxies. Learnable linear maps remain present; KUT is not a parameter-free architecture. Its claim is narrower and more auditable: the principal nonlinearity, transport, field, temperature, and readout modulation paths are tied to the same learned potential.

Contributions. This revised formulation makes four contributions. First, we give a precise code-level derivation chain for v13.3-minimal KUT. Second, we define a conservative flat-geometric limit proposition rather than claiming exact equality to a standard Transformer without qualifications. Third, we describe the three major geometric locks that reduce geometry-bypass routes while acknowledging that learnable Transformer-like linear maps remain. Fourth, we report current experimental observations on KUM-Lumina-0.6B as validation diagnostics, not as benchmark dominance claims.

2 Kähler Potential and Derived Quantities

2.1 Potential and shell chart

Let $r = |z|^2 = z_a^2 + z_b^2$. The potential is

$$K(r) = \frac{\alpha}{2} r^2 + \beta r, \qquad \alpha > 0, \quad \beta > 0. \tag{2}$$

The implementation represents the field by a bounded shell chart

$$\rho = \sqrt{v^2}\,\exp(\tanh s), \qquad z_a = \rho\cos\theta, \qquad z_b = \rho\sin\theta. \tag{3}$$

Thus $\rho/\sqrt{v^2} \in [e^{-1}, e]$. This chart is a numerical stability mechanism and an architectural constraint: it keeps field states near a learned vacuum shell without applying unstructured clipping.

2.2 Operational geometry

Differentiating equation (1) gives

$$\partial_z K = (\alpha r + \beta)\bar{z}, \qquad g = \partial_z \partial_{\bar{z}} K = 2\alpha r + \beta. \tag{4}$$

At the vacuum radius v^2, define

$$g_{\text{vac}} = 2\alpha v^2 + \beta. \tag{5}$$

The v13.3-minimal implementation uses the following operational geometric bundle:

$$g = 2\alpha r + \beta, \tag{6}$$

$$\Gamma = \frac{2\alpha}{g}, \tag{7}$$

$$\kappa = \frac{2\alpha\beta}{g^2}, \tag{8}$$

$$\omega = \sqrt{\frac{g_{\mathrm{vac}}}{v^2}}, \tag{9}$$

$$m = \sqrt{\frac{g}{g_{\mathrm{vac}}}}. \tag{10}$$

Here Γ and κ should be read as code-level connection and curvature proxies induced by the radial Kähler metric, rather than as a complete tensorial treatment of Kähler curvature. This convention is deliberate: the architecture requires deterministic scalar modulation fields, not full curvature tensors.

2.3 Probability side in v13.3-minimal

Earlier theory-complete variants included an explicit Boltzmann factor $\exp(-\Delta K/2\hbar)$. The v13.3-minimal branch removes this factor by an Occam reduction. The probability side enters operationally through two quantities:

$$\hbar_{\mathrm{eff}} = \frac{\pi}{\bar{\omega}}, \qquad m = \sqrt{g/g_{\mathrm{vac}}}, \tag{11}$$

where $\bar{\omega}$ is the shared bath frequency used by the layer integrator and output temperature. Thus the section amplitude m is better described as a metric probability-amplitude surrogate, not as a literal Boltzmann weight. The identity

$$\hbar_{\mathrm{eff}}\,\bar{\omega} = \pi \tag{12}$$

therefore holds by definition. The empirical question is not whether this identity appears, but whether training stabilizes $\bar{\omega}$ near the chosen boundary value

$$\omega^* = \frac{\pi}{e}. \tag{13}$$

3 Architecture

3.1 Overview

KUM-Lumina-0.6B uses the Qwen3-0.6B-scale backbone dimensions: hidden size 1024, 28 layers, 16 attention heads, 8 key-value heads, and complex field rank $R = 128$. The architecture consists of token embeddings, a field initializer $h \mapsto (s, \theta)$, repeated minimal Kähler layers, a final Riemannian normalization, and a section-flow readout.

Each layer uses one Kähler bundle evaluation at the layer start. Attention is the sole token-mixing mechanism; v13.3-minimal removes the retarded propagator, wave/Laplacian force, and explicit Boltzmann prior used in earlier comparison branches.

3.2 Kähler attention as transported inner product

Let the projected query and key be split into real and imaginary halves,

$$q = (q_a, q_b), \qquad k = (k_a, k_b). \tag{14}$$

The expanded Kähler kernel is

$$H + \Gamma A = q_a k_a + q_b k_b + \Gamma(q_a k_b - q_b k_a). \tag{15}$$

Because Γ is query-position local in v13.3, this expression is implemented by transporting the query before a single dot product:

$$q'_a = q_a - \Gamma q_b, \qquad q'_b = q_b + \Gamma q_a, \tag{16}$$

so that

$$\langle q', k \rangle = H + \Gamma A. \tag{17}$$

The attention score is

$$\mathrm{score}_{ij} = \frac{\pi}{\hbar_{\mathrm{eff}}} \frac{\langle q'_i, k_j \rangle}{\sqrt{d}}. \tag{18}$$

This form uses one standard $QK^\top$ multiplication and is compatible with scaled-dot-product attention kernels. In the flat limit $\Gamma \to 0$, equation (18) becomes ordinary scaled dot-product attention up to a constant temperature.

3.3 Residual transport

The residual state is not added in a purely flat form. Before each update, the hidden state is transported by a scalar connection factor

$$\mathrm{PT}_{\Gamma,dt}(h) = \exp(-\Gamma dt)\, h. \tag{19}$$

The attention update has the form

$$h \leftarrow \mathrm{PT}_{\Gamma,dt}(h) + \mathrm{Attn}_K(h), \tag{20}$$

and analogous transport is applied before the field-to-hidden and feed-forward updates. This is a scalar operational transport, not a full vector-bundle parallel transport map. It is nevertheless derived from the same connection proxy equation (7).

3.4 Field evolution

The field evolves only under the nonredundant local forces retained in the Occam branch:

$$F_{\mathrm{SSB}} = -\kappa_{\mathrm{vac}}(r - v^2)z, \tag{21}$$

$$F_{\mathrm{Calabi}} = -\omega(\kappa - \bar{\kappa})z. \tag{22}$$

Here $\kappa_{\mathrm{vac}} = 2\alpha\beta/g^2_{\mathrm{vac}}$. The sum of these forces is projected to the shell-chart coordinates (s, θ) and integrated with

$$dt = \frac{1}{L\bar{\omega}} \cdot c_{dt}, \tag{23}$$

where L is the number of layers and c_{dt} is the configured time-scale factor. The wave/Laplacian force and retarded propagator are omitted in v13.3-minimal because attention already supplies global token mixing.

3.5 Field-hidden coupling

After field evolution, the new field coordinates are weakly injected into the hidden state:

$$h \leftarrow \mathrm{PT}_{\Gamma,dt}(h) + dt\, W_{\mathrm{from}}[z_a^{\mathrm{new}}, z_b^{\mathrm{new}}]. \tag{24}$$

The current implementation initializes W_{from} near zero, with standard deviation 10^{-4}, rather than exactly zero. This gives geometry a weak forward path at initialization while avoiding large destabilizing injections.

3.6 Covariant gated MLP

The feed-forward block operates in a metric-scaled chart:

$$\mathrm{MLP}_K(h) = \frac{1}{\sqrt{g_h}} W_{\mathrm{down}} \Big[\tanh(W_{\mathrm{gate}}(h\sqrt{g_h})) \odot W_{\mathrm{up}}(h\sqrt{g_h}) \Big], \tag{25}$$

where g_h denotes the token-wise averaged metric ratio g/g_{vac} broadcast to the hidden dimension. Unlike an optional conditioning feature, g_h is required by the module interface. When g_h is constant, equation (25) reduces to a standard gated feed-forward block up to constant rescaling.

3.7 Section-flow readout

Project the hidden state to real and imaginary section coordinates $h_{\mathrm{re}}, h_{\mathrm{im}} \in \mathbb{R}^R$. The state section is

$$\Psi_a = m(\cos\theta\, h_{\mathrm{re}} - \sin\theta\, h_{\mathrm{im}}), \tag{26}$$
$$\Psi_b = m(\sin\theta\, h_{\mathrm{re}} + \cos\theta\, h_{\mathrm{im}}), \tag{27}$$

with $m = \sqrt{g/g_{\mathrm{vac}}}$. The flow section is a multi-timescale causal difference

$$\dot{\Psi}_t = \sum_{k\in\{1,2,4\}} \mathrm{Norm}(\Psi_t - \Psi_{t-k}), \tag{28}$$

where unavailable past positions are handled by left replication. The output logits are

$$\mathrm{logits}_t = \frac{\mathrm{Re}\langle \Phi, \Psi_t\rangle + \mathrm{Re}\langle \Xi, \dot{\Psi}_t\rangle}{\hbar_{\mathrm{eff}}}. \tag{29}$$

The same measurement basis Ξ is shared across flow scales, so the multi-timescale extension adds no new vocabulary-sized parameter matrix.

3.8 Loss

The training objective is

$$\mathcal{L} = \mathrm{CE}(\mathrm{logits}, y) + \left(\log \frac{\bar{\omega}}{\omega^*}\right)^2, \qquad \omega^* = \frac{\pi}{e}. \tag{30}$$

There is no auxiliary geometric alignment loss and no trainer-side flow loss in v13.3-minimal. Geometry is shaped through the cross-entropy path and the frequency boundary condition.

4 Flat-Geometric Limit

Proposition 1 (Controlled flat-geometric reduction)**.** *Let $\alpha \to 0$ with β and v^2 fixed. Then the K-derived modulation fields satisfy*

$$g \to \beta, \qquad \Gamma \to 0, \qquad \kappa \to 0, \qquad m = \sqrt{g/g_{\mathrm{vac}}} \to 1, \qquad \hbar_{\mathrm{eff}} \to \pi\sqrt{v^2/\beta}. \tag{31}$$

Consequently, Kähler attention reduces to standard scaled dot-product attention with constant temperature, residual transport reduces to ordinary residual propagation, and the covariant MLP reduces to a standard gated MLP up to constant scaling. If the explicitly geometric side channels—field-to-hidden injection and section-flow readout—are additionally closed, the architecture reduces to a Transformer-family block with fixed output temperature.

Proof. The limits for g, Γ, κ, m, and $\hbar_{\text{eff}}$ follow directly from equations (6) to (10). Substituting $\Gamma = 0$ in equation (16) gives $q' = q$, so the Kähler attention score becomes a constant-temperature scaled dot product. The transport factor $\exp(-\Gamma dt)$ tends to 1, so transported residual propagation becomes flat residual propagation. Finally, the metric factor in equation (25) is constant and cancels between input and output scaling. The remaining chart and section-flow mechanisms are not standard Transformer operations; closing them gives the stated Transformer-family reduction. □

Remark 1. *The proposition is intentionally weaker than an exact-recovery theorem. The implemented model contains a shell chart and a section-flow readout even when α is small. Thus "flat limit" should be understood as a controlled reduction of the K-derived modulation fields, not as an unconditional identity with a standard pretrained Transformer implementation.*

5 Geometry Locks and Remaining Shortcuts

The architecture reduces geometry-bypass routes through three major locks:

Lock 1: **Transport lock.** Residual propagation depends on $\Gamma = 2\alpha/g$.

Lock 2: **Covariant MLP lock.** Feed-forward computation uses g/g_{vac} on both input and output charts.

Lock 3: **Section-readout lock.** Logits depend on $m = \sqrt{g/g_{\text{vac}}}$ and on $\hbar_{\text{eff}} = \pi/\bar{\omega}$.

These locks mean that every principal output path is K-modulated. However, the model still contains learnable linear maps in the attention projections, MLP, field initializer, and readout bases. It is therefore more precise to say that KUT substantially reduces geometry-bypass routes, rather than that no computational path can reduce cross-entropy without changing α.

6 Experimental Observations

6.1 Setup and data-window discipline

Current KUM-Lumina-0.6B runs initialize embeddings and major network weights from a Qwen3-0.6B-scale backbone, then train the full model including the geometric parameters. FineMath stages use disjoint token windows: for a given run, the stream first skips a configured number of tokens, then consumes a training window, then carves a non-overlapping evaluation window after the training window. This avoids using the tail of the training window as evaluation data.

6.2 Training dynamics

Stage	Data	Tokens	Best PPL	α	$\bar{\omega}$	Top-1
1	WikiText-103	100M	96.0	0.499	1.256	0.356
2	FineMath window 1	100M	308.0	0.541	1.336	0.286
3	FineMath window 2	100M	53.7	0.572	1.245	0.441
4	FineMath window 3	150M	43.4	0.654	1.237	0.482

Table 1: Representative staged training diagnostics from current KUM-Lumina-0.6B runs. Values are architecture-validation diagnostics rather than benchmark claims.

The main observations are:

(i) α increases substantially across stages, indicating that cross-entropy training can strengthen the learned curvature channel without an auxiliary geometric loss.

(ii) The identity $\hbar_{\text{eff}}\bar{\omega} = \pi$ is enforced by definition; the empirical diagnostic is the stabilization of $\bar{\omega}$ near $\omega^* = \pi/e$.

(iii) The logit-stasis diagnostic decreases in current runs, indicating that consecutive token distributions become less degenerate during training.

6.3 Generation diagnostics

Greedy generation from current checkpoints produces coherent short completions in both natural-language and mathematical prompts. We also report a repetition diagnostic rep1, defined as the fraction of generated tokens equal to the immediately preceding generated token. A value near zero indicates absence of immediate one-token repetition, not a general guarantee of long-range diversity.

Prompt	Representative greedy completion
Albert Einstein was a	member of the National Register of Historic Places.
Theorem: For all integers n,	there exists a positive integer n such that $n^2 - 1$ is a factor of the product of the integers.
Consider the function $f(x) =$	$2x^3 + 10x^2 + 12x + 12$. Find the value of $f(x)$ if x is a positive real number.

Table 2: Representative qualitative generations. These examples demonstrate nondegenerate decoding, not state-of-the-art language modeling.

6.4 Section-readout ablation

An operational ablation replacing the section readout by a standard linear head produced degenerate greedy output in current tests, while the section readout produced coherent short completions. This supports the claim that the geometric readout is functionally important in this implementation. The stronger statement that geometry is mathematically necessary for all possible language generation should not be inferred from this ablation alone.

6.5 Causal consistency diagnostic

Causality is checked by comparing teacher-forced single-pass predictions with incremental autoregressive predictions under no-cache decoding. In the reported diagnostic, all tested positions matched. This is evidence against token-future leakage in the tested path; it is not a proof of all possible implementation paths unless the diagnostic is run after each architectural change.

7 Related Work

KUT is related to, but distinct from, several lines of work. Scaled dot-product attention originated in the Transformer architecture [1]. RMS normalization [2], gated activations [3], and rotary embeddings [4] are standard components that KUT reinterprets through a shared geometric modulation framework. Geometric deep learning [5] studies neural computation on structured domains, but typically does not derive Transformer internals from a single potential. Hyperbolic and Riemannian neural methods optimize or represent data on curved spaces, whereas KUT learns the scalar geometry that modulates its own computation.

Earlier KUT variants used explicit geometric alignment losses, Hermitian or symplectic attention kernels, trainer-side one-pass regularizers, retarded propagators, and wave/Laplacian field forces. The v13.3-minimal branch removes these redundant or weakly validated channels and retains only the Kähler attention, transported residual, SSB/Calabi field dynamics, covariant MLP, section-flow readout, and CE-plus-anchor loss.

8 Discussion

8.1 Meaning of increasing α

The parameter α controls the quartic part of equation (1). In the controlled flat-geometric limit $\alpha \to 0$, K-derived modulation weakens. The observed increase of α during training is therefore evidence that the optimizer finds curvature useful in the current architecture. A stronger causal claim requires controlled ablations, such as freezing $\alpha = 0$, freezing α at initialization, or replacing K-derived modulation by learned scalar gates with matched parameter counts.

8.2 Role of the shell chart

The shell chart equation (3) prevents radial divergence and bounds the metric ratio in a compact range determined by $\rho/\sqrt{v^2} \in [e^{-1}, e]$. This is one reason the minimal branch does not require an explicit Boltzmann factor $\exp(-\Delta K/2\hbar)$: the field is already restricted to a near-vacuum annulus.

8.3 Quantum-mechanical analogy

The section readout resembles a measurement of a complex state: phase θ rotates the projected hidden coordinates, $m = \sqrt{g/g_{\text{vac}}}$ weights the section amplitude, and the readout basis measures $\mathrm{Re}\langle \Phi, \Psi \rangle$. This analogy is structurally useful, but it should not be read as a claim that the model implements physical quantum mechanics. Likewise, $\hbar_{\text{eff}}\bar{\omega} = \pi$ is a designed quantization identity in the implementation, while stabilization of $\bar{\omega}$ near π/e is the empirical phenomenon.

8.4 Limitations

KUM-Lumina-0.6B does not match the performance of fully pretrained models of comparable size. The current evidence validates architectural functionality, causal cleanliness under tested diagnostics, and nondegenerate generation. It does not yet establish benchmark competitiveness, scaling laws, or universal necessity of Kähler geometry for language modeling.

9 Conclusion

We introduced a conservative formulation of the Kähler Unified Transformer v13.3-minimal branch. A single quartic Kähler potential generates the metric, connection proxy, curvature proxy, field forces, temperature, section amplitude, and measurement path used by the model. The architecture is best understood as a K-modulated Transformer-family model whose principal output paths are geometrically constrained.

The current experiments support four bounded conclusions: coherent short-form generation is possible; the section readout is operationally important in the tested implementation; cross-entropy training can increase the learned curvature parameter α without auxiliary geometric loss; and tested teacher-forced/incremental predictions agree under the causal diagnostic. These findings justify further scaling and ablation studies, while leaving open the stronger questions of benchmark competitiveness and exact theoretical equivalence to standard Transformers.

A Derivation Chain

Listing 1: Operational derivation chain in KUT v13.3-minimal.

```
K(z,zbar) = (alpha/2)|z|^4 + beta|z|^2
|-- Geometry
| |-- g = 2 alpha r + beta -> RiemannianNorm, covariant MLP
| |-- Gamma = 2 alpha / g -> attention transport, residual transport
| |-- kappa = 2 alpha beta / g^2 -> SSB and Calabi field forces
| `-- omega = sqrt(g_vac / v^2) -> dt and hbar = pi / omega
|-- Probability / temperature
| |-- sqrt(g/g_vac) -> section amplitude surrogate
| `-- hbar = pi / omega -> readout and attention temperature
|-- Physics
| |-- F_SSB = -kappa_vac (r-v^2) z
| `-- F_Calabi = -omega (kappa-kappa_bar) z
`-- Information
    |-- Psi = sqrt(g/g_vac) R(theta) proj(h)
    |-- flow = sum_k Norm(Psi[t]-Psi[t-k]), k in {1,2,4}
    `-- logits = (Re<Phi,Psi> + Re<Xi,flow>) / hbar
```

B Hyperparameters

Parameter	Value	Comment
hidden size	1024	Qwen3-0.6B-scale backbone
number of layers	28	Qwen3-0.6B-scale depth
attention heads / KV heads	16 / 8	grouped-query attention configuration
complex rank R	128	field and section rank
init α_{raw}	-3.0	low-curvature initialization after softplus
init β_{raw}	0.0	softplus(0) $= \log 2$ before floor
init v^2_{raw}	inverse-softplus(1.0)	unit vacuum initialization
RoPE theta	10^6	long-context rotary scale
flow scales	$(1, 2, 4)$	multi-timescale causal flow readout
network learning rate	3×10^{-4}	current training setting
geometric learning rate	10^{-3}	current training setting
optimizer	AdamW	$\beta_1 = 0.9$, $\beta_2 = 0.95$ in current trainer
sequence length	512	single-GPU memory-limited setting

Table 3: Representative hyperparameters for the current KUT v13.3-minimal runs.

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